Omar Ait-Mlouk

AI Engineer | Python Developer

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AI Engineer and Software Developer with 4+ years of experience building Python applications, ML models, and AI-integrated tools. Currently researching Federated Learning and Retrieval- Augmented Generation (RAG) for healthcare decision support. Experienced in LLM integration, NLP, computer vision, and production-ready AI pipelines.

Work Experience

AI Engineer (Part-Time) *Jul 2024 — Present*
AINNOVATEX Marrakech, Morocco

- Built LLM-powered document processing system using OpenAI and Claude APIs, automating extraction of structured data from 500+ unstructured documents per week
- Engineered RAG pipeline with LangChain, FAISS, and sentence embeddings, achieving 92% retrieval accuracy on internal knowledge base of 10K+ documents
- Deployed computer vision model for automated visual inspection, reducing manual review time by 60%
- Developed NLP-based classification tool processing Arabic and French text, serving 3 internal teams

Python Developer *Jun 2023 — Jun 2024*
Batteria Casablanca, Morocco

- Engineered automated data pipelines processing 10K+ daily transactions across sales, inventory, and HR modules
- Built REST API layer integrating 5+ third-party services, reducing manual data entry by 70%
- Designed PostgreSQL database schemas and optimized queries, improving report generation speed by 3x

Software Engineer *Jan 2021 — May 2023*
AINNOVATEX Marrakech, Morocco

- Architected full company platform from scratch using Python/Odoo, serving as the core business system
- Built maintenance management module (GMAO) that digitized 100% of maintenance workflows
- Developed mobile app with FastAPI backend enabling real-time field access to business-critical data

Projects & Research

- **Federated Learning–Enabled RAG for Healthcare** *(Master's Thesis)* — Designing a privacy-preserving clinical decision support system combining federated learning with RAG across distributed medical institutions. Stack: PyTorch, LangChain, FAISS, Flower
- **AI-Powered Code Review Assistant** — Built a tool that analyzes pull requests using LLMs and embeddings, providing automated code suggestions and detecting potential bugs across Python and JavaScript codebases
- **Multilingual NLP Pipeline** — End-to-end text classification and entity extraction pipeline for Arabic/French documents using Hugging Face Transformers and spaCy

Education

Master's Degree in Artificial Intelligence and Applications *2024 – 2026*
Ibn Zohr University, Morocco
Key Coursework: Deep Learning, NLP, Computer Vision, Generative AI, Federated Learning, Multi-Agent Systems

Bachelor's Degree in Software Engineering *2023*
UPM Marrakech, Morocco

Technical Skills

AI / ML	PyTorch, TensorFlow, Scikit-learn, Hugging Face, LangChain, FAISS, RAG, Federated Learning, Deep Learning, NLP, Computer Vision, Generative AI, Prompt Engineering, MLOps, Model Fine-Tuning
Programming	Python, SQL, JavaScript
Data Engineering	Pandas, NumPy, Apache Spark, PostgreSQL, MongoDB, Power BI
Tools & DevOps	Git, Docker, FastAPI, Django, Linux, REST APIs, Agile/Scrum
Languages	Arabic (Native) English (Fluent) French (Intermediate)